

The following is a short description of my project, Palm Pilot.

# The Palm Pilot

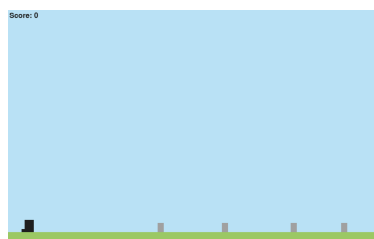
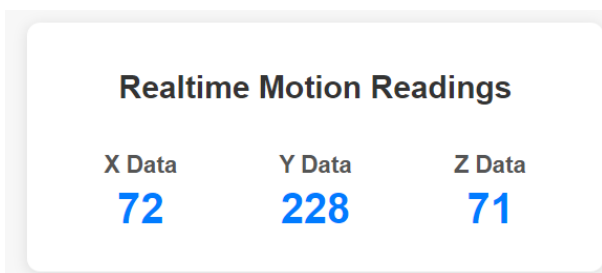
## Abstract

The Palm Pilot is a wearable glove designed to assist disabled individuals in their daily lives. Its features include audible notifications for nearby objects, navigation directions, continuous monitoring of hand movements and heart rate (updated periodically to a website), and the ability to control household appliances through hand gestures. Additionally, it can be used in hospital wards, allowing nurses to monitor multiple patients via a web interface. The glove also provides entertainment by enabling a button to wirelessly control games like Flappy Bird and the Chrome dinosaur game on a computer screen or an external display connected to the central device. It can also control a laptop cursor, though it is limited to basic pointer movement.

## Working Mechanism

The glove employs radio communication to send signals to a central device located in the user's home. Pre-programmed hand movements trigger specific actions, with gestures captured using motion and flex sensors. For example, raising the palm turns on a light. Ultrasonic and other sensors map the surroundings, alerting the user if an object is in close proximity. The pitch and intensity of the alert sound vary with the object's distance from the glove. Data is periodically sent to a database (once every 30 seconds) about the hand movement and the heart rate of the user. The glove achieves this by connecting to the internet, so it requires an internet connection in order to update values. These values are also updated on a website, which dynamically changes.

For the purpose of playing the game, radio communication is used to play on the central device and bluetooth connectivity to play on the laptop. As soon as the button on the glove is pressed, a value is sent through both the sources and the characters jump. The motion of the hand in the XY plane is also used to map a cursor to a respective pixel on the screen. It is achieved through python's PyAutoGUI library.



```
// Finger sensor
Firebase.config({apiKey, authDomain});
firebase.auth().signInWithEmailAndPassword('1234567890', '1234567890');

// Serial
const Serial = require('serialport');
const port = new Serial({
  path: '/dev/tty.usbmodem14101',
  baudRate: 9600
});

// Data
const data = {
  x: 0,
  y: 0,
  z: 0,
  heartRate: 0,
  distance: 0,
  time: 0,
  score: 0,
  game: 'Flappy Bird'
};

// Send data to Firebase
const firebase = require('firebase');
const firebaseRef = firebase.database().ref('data');

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```